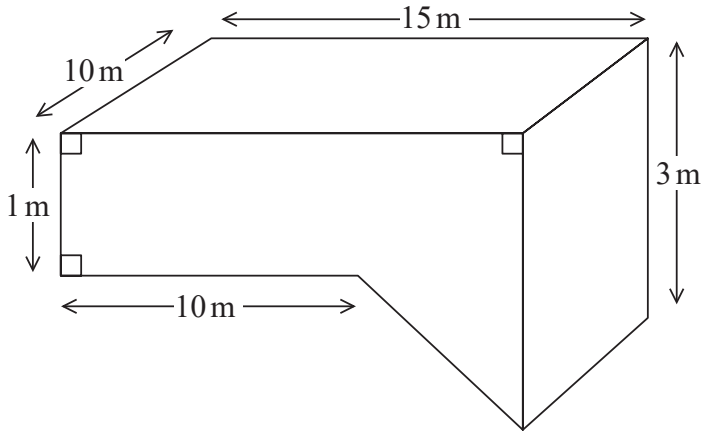
21



The diagram shows a swimming pool.
The swimming pool is in the shape of a prism.
The swimming pool is filled with water at a rate of 5 litres per second.

Jeremy has 10 hours to fill the swimming pool.
$1\,m^3 = 1000$ litres.

Will he completely fill the swimming pool in 10 hours?
You must show all your working.

Side cross-section: rectangle15 × 1 plus triangle.

Triangle base = 15 – 10 = 5 m; height = 3 – 1 = 2 m.

Area = 15 × 1 + ½ × 5 × 2 = 20 m^2.

1 mark
Cross-sectional area

Pool volume = 20 × 10 = 200 m^3.

1 mark
Prism volume

200 m^3 = 200000 litres.

1 mark
Convert to litres

Time = 200000 ÷ 5 = 40000 seconds.

1 mark
40000 seconds

10 hours = 10 × 60 × 60 = 36000 seconds.

No: 40000 > 36000; he needs more than10 hours.

1 mark
Compare same units

(Total for Question 21 is 5 marks)